# Junho Peter Whang

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## Academic Positions

**Seoul National University**, Seoul, Korea — Aug. 2021–present
Assistant Professor, Department of Mathematical Sciences

**Princeton University**, Princeton, NJ, USA — Sep. 2025–Jun. 2026
Visiting Fellow, Department of Mathematics (sabbatical)

**Massachusetts Institute of Technology**, Cambridge, MA, USA — Jul. 2018–Jun. 2021
C.L.E. Moore Instructor

## Education

**Princeton University**, Princeton, NJ, USA — 2018
Ph.D. in Mathematics. Advisor: Peter Sarnak.
Thesis: *Diophantine Analysis on Moduli of Local Systems.*

**Queen's University**, Kingston, ON, Canada — 2013
B.Sc. in Mathematics, Honours with Distinction.

## Research Interests

Number theory (Diophantine analysis), arithmetic geometry, topology, and dynamics.

## Publications and Preprints

1. *On indefinite integral ternary quadratic forms.*
   With Alexander Gamburd, Amit Ghosh, and Peter Sarnak. Preprint, arXiv:2603.05849.
2. *Degree gaps in algebraic interpolations.*
   Bull. Korean Math. Soc. **63** (2026), no. 1, 227–234.
3. *A reciprocity law on the one-holed torus.*
   J. Korean Math. Soc. **62** (2025), no. 6, 1525–1538.
4. *On finite categories of algebraic varieties.*
   J. Algebra **681** (2025), 206–217.
5. *Stokes matrices and exceptional isomorphisms.*
   With Yu-Wei Fan. Math. Ann. **390** (2024), 4041–4086.
6. *On periodic orbits of polynomial maps.*
   Preprint, arXiv:2305.13529.
7. *Surface group representations in* $\mathrm{SL}_2(\mathbb{C})$ *with finite mapping class orbits.*
   With Indranil Biswas, Subhojoy Gupta, and Mahan Mj. Geom. Topol. **26** (2022), no. 2, 679–719.
8. *The rank two p-curvature conjecture on generic curves.*
   With Anand Patel and Ananth Shankar. Adv. Math. **386** (2021).
9. *Arithmetic of curves on moduli of local systems.*
   Algebra Number Theory **14** (2020), no. 10, 2575–2605.
10. *Nonlinear descent on moduli of local systems.*
    Israel J. Math. **240** (2020), 935–1004.

11. *Global geometry on moduli of local systems for surfaces with boundary.*
    Compos. Math. **156** (2020), no. 8, 1517–1559.
12. *The uncertainty principle and a generalization of a theorem of Tao.*
    With M. Ram Murty. Linear Algebra Appl. **437** (2012), no. 1, 214–220.
13. *Another proof of the infinitude of the prime numbers.*
    Amer. Math. Monthly **117** (2010), no. 2, 181.

## Grants and Research Funding

**Samsung Science and Technology Foundation** 2022–2027
*Analysis of Diophantine Problems with Nonlinear Symmetry.* SSTF-BA2201-03.
Principal Investigator. Total project award: KRW 755 million.

**National Research Foundation of Korea** 2025–2028
*Dynamics and Arithmetic Structures on Manifolds (DASOM).* RS-2025-02293115.
Co-investigator. Total project award: KRW 500 million.

## Selected Honors and Awards

**Teaching and Learning Award**, Massachusetts Institute of Technology 2021
**Bershadsky Family Fellowship in Mathematics**, Princeton University 2017–2018
**NSERC Postgraduate Scholarship PGS D** 2014–2017

## Research Supervision

### Ph.D. students

- Seong Youn Kim, 2022–present; co-advised with Gye-Seon Lee.
- Young Joon Ley, 2022–present.
- Eungchan Kim, 2025–present.
- Jihye Oh, 2025–present.

### Master's students

- Juhan Kim, M.S. 2023.
  Thesis: *Compactification of Character Varieties via Triangulation on Surfaces.*
- Eunju Shin, M.S. 2024.
  Thesis: *Nonlinear Reduction for Two Types of Generalized Markoff Equations.*
- Jihye Oh, M.S. 2025.
  Thesis: *Decidability of Mutation Equivalence for Three-Vertex Quivers.*

## Teaching

**Seoul National University** 2021–present
Graduate: Algebra, Algebraic Geometry, Commutative Algebra, Diophantine Geometry.
Advanced Topics in Number Theory.
Undergraduate: Linear Algebra, Introduction to Diophantine Analysis, Analytic Number Theory.

**Massachusetts Institute of Technology** 2018–2021
Introduction to Arithmetic Geometry (18.782).